PARTIAL FRACTIONS

If we consider the integral $\int \frac{f(x)}{x^3+ax^2+cx+d}\,dx$ and the denominator can be factorized in linear or quadratic functions, then we apply partial fraction method.

Example 1

Evaluate the integrals:

(a) $\int \frac{2x-3}{(x-1)(x-2)(x+3)}\,dx$ (b) $\int \frac{x}{x^2-5x+6}\,dx$

(c) $\int \frac{x^2-2x}{(2x+1)(x^2+1)}\,dx$ (d) $\int \frac{8-x}{(x-2)^2(x+1)}\,dx$

Solution

Cover up rule:

(a) $\frac{2x-3}{(x-1)(x-2)(x+3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x+3}$; $\begin{cases} \underset{(x=1)}{A} = \frac{-1}{(-1)(4)} = \frac{1}{4} \\ \underset{(x=2)}{B} = \frac{1}{1(5)} = \frac{1}{5} \\ \underset{(x=-3)}{C} = \frac{-9}{-4(-5)} = -\frac{9}{20} \end{cases}$

$$\int \frac{2x-3}{(x-1)(x-2)(x+3)}\,dx = \frac{1}{4}\ln(x-1) + \frac{1}{5}\ln(x-2) - \frac{1}{20}\ln(x+3) + C$$

(b) $\int \frac{x}{x^2-5x+6}\,dx = \int \frac{x}{(x-2)(x-3)}\,dx$; Coverp $\frac{x}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$ $\begin{cases} \underset{(x=2)}{A} = \frac{2}{-1} \\ \underset{(x=3)}{B} = \frac{3}{1} \end{cases}$

$$= -2\ln(x-2) + 3\ln(x-3) + C$$

Partial fractions:

(c) $\frac{x^2-2x}{(2x+1)(x^2+1)} = \frac{A}{2x+1} + \frac{Bx+C}{x^2+1}$

Compare Coefficients

ie $x^2-2x = A(x^2+1) + (Bx+C)(2x+1) = (A+2B)x^2 + (B+2C)x + A + C$

Comparing Coefficients:

$A + C = 0$, ie $A = -C$, $B + 2C = -2$; $A + 2B = 1$

if $x = -\frac{1}{2}$; $\frac{1}{4} + 1 = \left(\frac{1}{4} + 1\right)A$ ie $A = 1$, $C = -1$,

$B = -2 + 2 = 0$;

So that

$$\int \frac{x^2 - 2x}{(2x+1)(x^2+1)}\,dx = \frac{1}{2}\ln(2x+1) - \tan^{-1}(x) + C$$

(d) Partial fractions:

$$\frac{8-x}{(x-2)^2(x+1)} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+1}$$

$$8 - x = A(x-2)(x+1) + B(x+1) + C(x-2)^2$$

$x = 2$: $6 = 3B$, $\therefore B = 2$

$x = -1$: $9 = 9C$, $\therefore C = 1$

$x = 0$: $8 = -2A + 2 + 4$, $\therefore A = -1$

So that:

$$\int \frac{8-x}{(x-2)^2(x+1)}\,dx = -\ln(x-2) - \frac{2}{\cancel{3}}(x-2)^{\cancel{-3}} + \ln(x+1) + D$$

$$= -\ln(x-2) - \frac{2}{x-2} + \ln(x+1) + D$$